JECRC UNIVERSITY JAIPUR
Session: 2025-26
Subject: Engineering Mathematics-II (DMA025A)
Course: B. Tech - II Semester
Tutorial Sheet – 11 (CO1)
UNIT-1

M.M:64

CO1: Use matrices, determinants and techniques for solving systems of linear equations in the different areas of Linear Algebra. Understand the definitions of Vector Space and its linear Independence. Solve Eigen value problems and apply Cayley Hamilton Theorem.

PART – A

(5x2=10)

A1 Complete the given statements:

a) Rank of a null matrix is_____________.

b) A system is said to be ____________if it has at least one set of solution.

c) The sum of the eigen values of a matrix A is equal to the _______of the matrix.

d) If 1,-2,3 are the eigen values of any square matrix A , then the eigen values of 3A are _______

A2 Show that the vectors (3,1,5) and (1,2,1) are linearly independent.

A3 For which value of x will be the matrix $\begin{bmatrix} 8 & x & 0 \\ 4 & 0 & 2 \\ 12 & 6 & 0 \end{bmatrix}$ becomes singular.

A4 Obtain the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 & -1 \\ -2 & -1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$ using Echelon form.

A5 Obtain the rank of the matrix $A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 3 & 4 & 0 & -1 \\ -1 & 0 & -2 & 7 \end{bmatrix}$ using Normal form

PART – B

(3x7=21)

B1 Using elementary row transformations, find the inverse of the matrix $A = \begin{bmatrix} 1 & 2 & 2 \\ -1 & 1 & 1 \\ 3 & -2 & 1 \end{bmatrix}$

B2 Discuss the consistency of the system and if consistent, solve the equation:

$$6x + y + z = -4$$
$$2x - 3y - z = 0$$
$$-x - 7y - 2z = 7$$

B3 For what values of k, the system of simultaneous equations

$$x + y + z = 1$$
$$2x + y + 4z = k$$
$$4x + y + 10z = k^2$$

have a solution and solve them completely in each case.

<u>PART – C</u>

(3x11=33)

C1 Find the eigen value and eigen vectors of the matrix $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

C2 Find the eigen value and eigen vectors of the matrix $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$

C3 State Cayley Hamilton theorem and verify it for the matrix $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ and hence find A^{-1}.

Answers:

A1 -----	A2 To prove	A3 4	A4 2	**A5** 2

B1 $A^{-1} = \frac{1}{9}\begin{bmatrix} 3 & -6 & 0 \\ 4 & -5 & -3 \\ -1 & 8 & 3 \end{bmatrix}$	B2 Consistent , $x = -1, y = -2, z = -4$	B3 K= 1,2
C1 $2, -1, -1$ $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$	C2 $-1,1,4$ $\begin{bmatrix} -6 \\ -2 \\ 7 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}$	C3 $1/4\begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$